LESSON 10.6

Circumference and arc length

Date _____

Learning intention We are learning to find circumferences, arc lengths and perimeters of curved shapes.

Success criteria
- ☐ I can use C = πd or C = 2πr to find a circumference
- ☐ I can find an arc length: $\frac{\theta}{360} \times 2\pi r$
- ☐ I can find the perimeter of sectors and composite shapes

NOTES — Key ideas for this lesson

TEACHER COPY

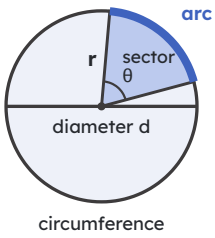
KEY TERMS

circumference (C): the distance around a circle
radius (r): from the centre to the edge
diameter (d): across through the centre; d = 2r
arc: part of the circumference
sector: a slice of a circle between two radii
θ: the angle at the centre of a sector
perimeter: the distance around the outside of a shape

FORMULAS

C = πd = 2πr
Arc length = $\frac{\theta}{360} \times 2\pi r$
Semicircle: P = ½πd + d
Quadrant: P = ¼ × 2πr + 2r
Sector: P = arc + 2r

STEPS 1 r = d ÷ 2 2 write the formula 3 substitute (π key) 4 round to 1 d.p. + unit



circumference

Level 1 · Getting started

Easy · 12 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Find each circumference or perimeter, correct to one decimal place.

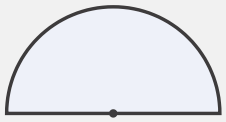
a Circumference



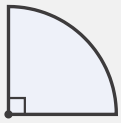
b Circumference



c Perimeter



d Perimeter



YOU DO On your own: Easy work down the box, one round at a time. Write each answer on its line.

12 QUESTIONS

REMEMBER C = πd = 2πr · Semicircle P = ½ × πd + d · Quadrant P = ¼ × 2πr + r + r · Use the π key.

ROUND 1 Questions 1–4: find the circumference of each circle with radius r, correct to one decimal place.

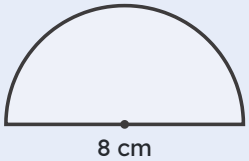
1. r = 3 cm
2. r = 7 m
3. r = 2.5 m
4. r = 1.5 km

ROUND 2 Questions 5–8: find the circumference of each circle with diameter d, correct to one decimal place.

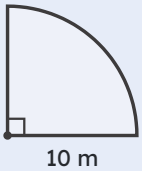
5. d = 8 cm
6. d = 15 m
7. d = 30 mm
8. d = 1.4 km

ROUND 3 Questions 9–12: find the perimeter of each shape, correct to one decimal place.

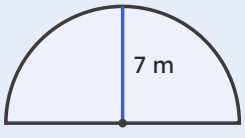
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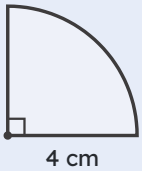
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11.



12.



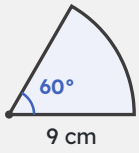
Level 2 · Building up

Medium · 16 questions

WE DO With your teacher: copy what they write on the board.

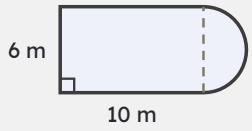
Example 2 **Give each answer correct to one decimal place.**

a Arc length



b Perimeter of the sector in part a

c Perimeter



d Radius of a circle with circumference 50 cm

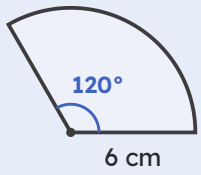
YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line.

16 QUESTIONS

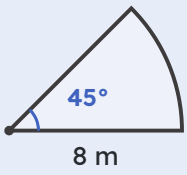
REMEMBER Arc length = $\frac{\theta}{360} \times 2\pi r$ (θ = the angle at the centre) · Sector P = arc + r + r · Composite: add the outside edges, not the dashed lines.

ROUND 1 Questions 1–4: find the perimeter of each sector, correct to one decimal place.

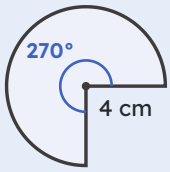
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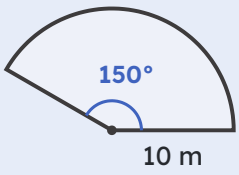
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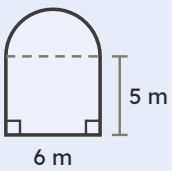


ROUND 2 Questions 5–8: find the perimeter of each shape, correct to one decimal place.

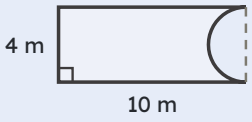
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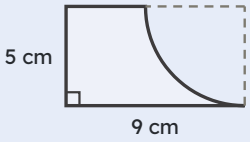
6



7 Cut-out: d = 4 m



8 Cut-out: r = 5 cm



ROUND 3 Questions 9–16: mixed. Read each one carefully: a tag shows an earlier lesson.

9 A wheel of diameter 64 cm turns 100 times. How far does it go, in m?

10 Find the perimeter of a rectangle 7.5 m by 4 m. 10.4

11 Convert 345 cm to m. 10.1

12 A round garden has a circumference of 18.85 m. Find its radius.

13 A kite has diagonals of 6 m and 5 m. Find its area. 10.5

14 Write the limits of accuracy of 4.5 m. 10.2

15 Find the arc length of a 120° sector with r = 3 m.

16 Find the percentage error of 20 cm, to the nearest cm. 10.3